



UC San Diego

JACOBS SCHOOL OF ENGINEERING

Electrical and Computer Engineering

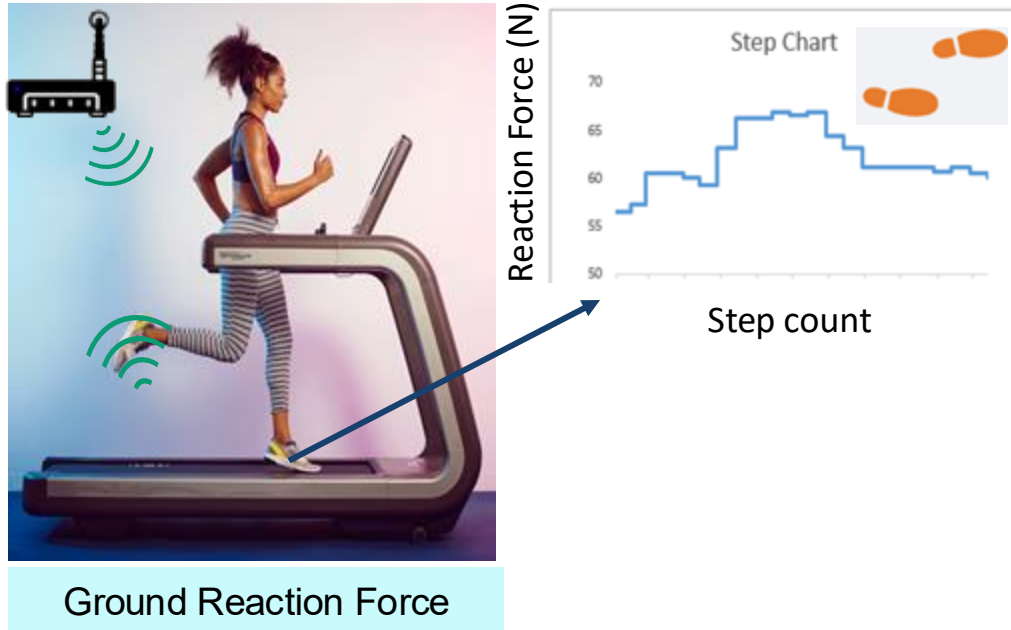


# *ZenseTag*: An RFID assisted Twin-Tag Single Antenna COTS Sensor Interface

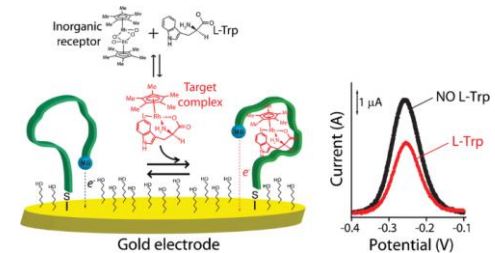
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Nagarjun Bhat\*, Agrim Gupta, Ishan Bansal, Harine  
Govindarajan, Dinesh Bharadia

# Ubiquitous sensing - Next wave of IoT



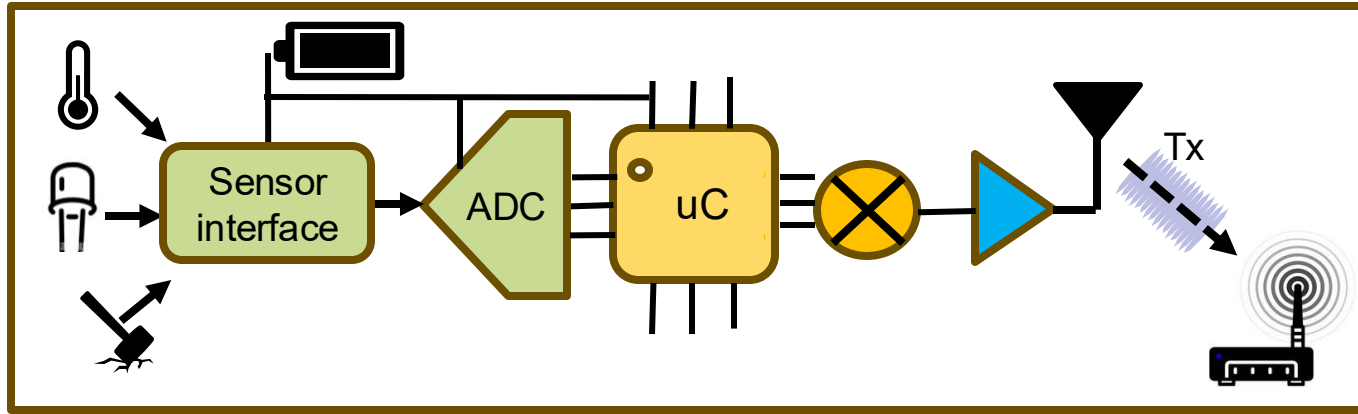
Sensing soil-moisture to automate irrigation



Biochemical sensors<sup>1</sup>

Sensors enable continuous data collection for ubiquitous sensing.

# Challenges of ubiquitous sensing: Batteries and bulk

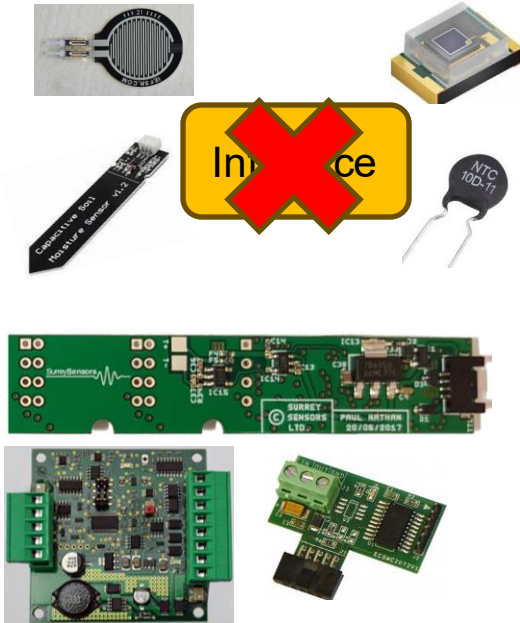


➤ Bulky Sensor interfaces with batteries!

- Need batteries/ energy harvesters.
- Rigid PCB with complex electronics to read sensors.

**Sensor interfaces are bulky, rigid and need batteries!**

# Challenges of ubiquitous sensing: Sensor Interface



- A zoo of sensors to choose from! But. . .
  - Every sensor outputs different voltage/ current!!
  - Each sensor needs a unique interface!
  - No universal sensor interface!

No universal interface for COTS sensors.

# Challenges of ubiquitous sensing: Radio infrastructure



- So many commercial radio systems available!
  - None of which readily support battery-free sensor interfaces!
  - Passive sensing needs SDRs and custom waveforms.

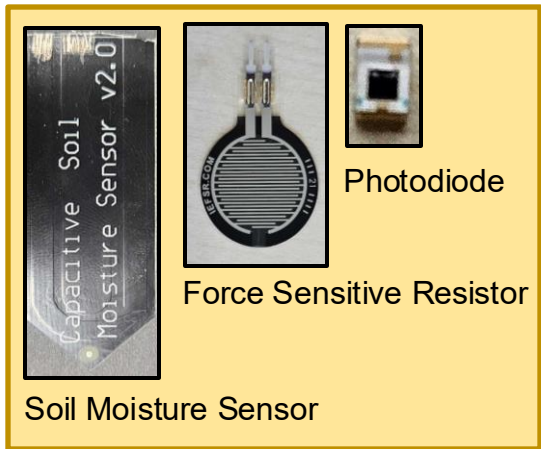
No passive interface compatible with existing radios.

# Current passive interfaces don't meet these needs:

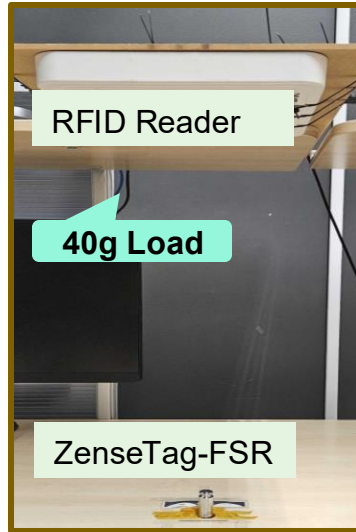
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| Requirements                    | Digital Interfaces                                                                | Analog Interfaces                                                                   | ZenseTag                                                                            |
|---------------------------------|-----------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| Universal Interface             |  |  |  |
| Compact / Simple form factor    |  |  |  |
| Compatible with existing radios |  |  |  |
| Reliable despite multipath      |  |  |  |
| Realtime sensing                |  |  |  |

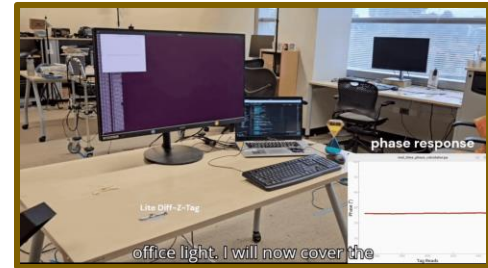
# ZenseTag: Contributions



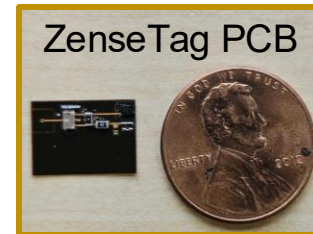
COTS Sensor Interface



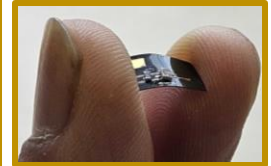
Battery-Free/ commercial  
RFID Compatible



Robust and Realtime



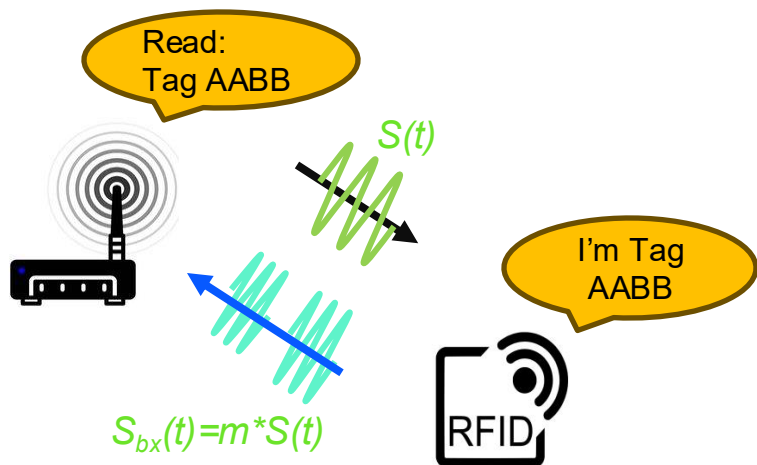
15x10mm



Compact/ flexible form factor

ZenseTag: Battery-free, reliable sensor interface for  
COTS sensors using commercial RFID.

# Can RFID tags enable battery-free sensing?

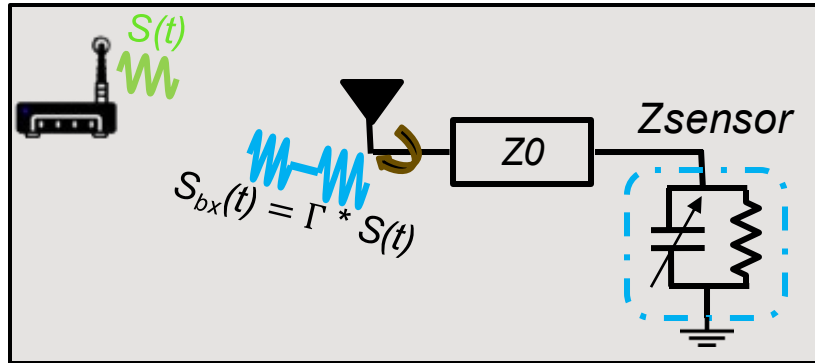


- RFID is a ubiquitous radio platform.
- RFID tags are passive/flexible and inexpensive
- RFID tags simply backscatter their digital ID.
- No inherent sensing capability.

Can we use the digital ID of Tags for sensing?

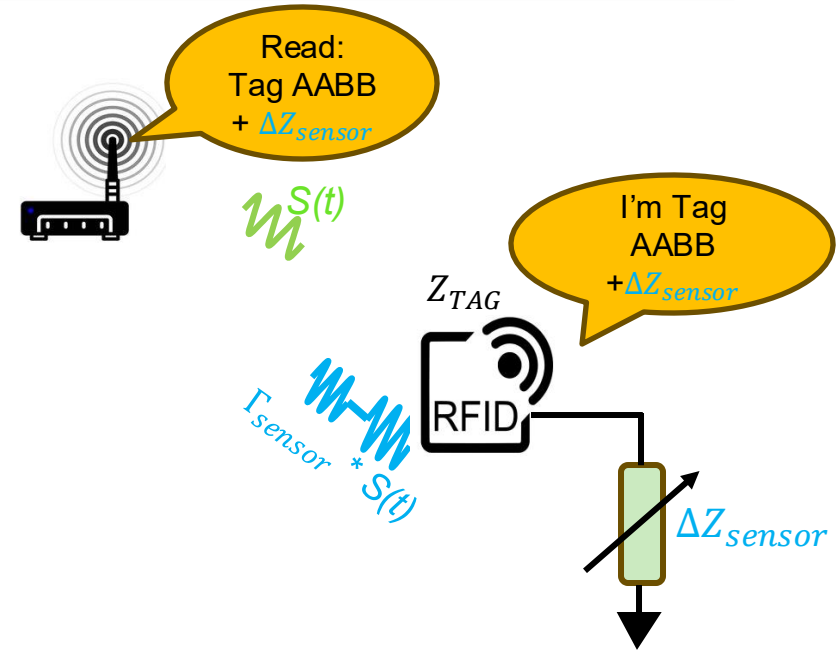
# How to modulate RFID signal with sensor output?

$$Z_{\text{sensor}} = \frac{V_{\text{sensor}}(f)}{I_{\text{sensor}}(f)}$$



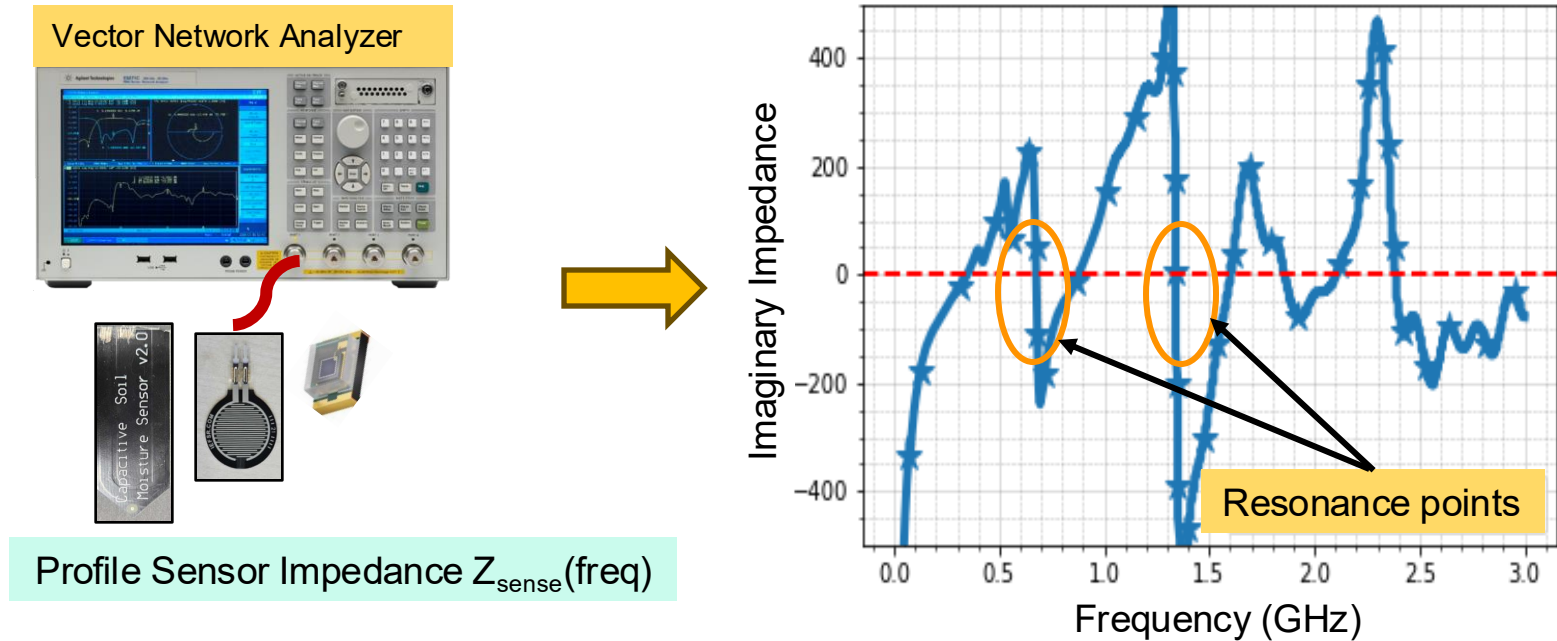
$$\Gamma = \frac{Z_{\text{sensor}} - Z_0}{Z_{\text{sensor}} + Z_0}$$

$$Z_0 \rightarrow Z_{\text{TAG}}$$



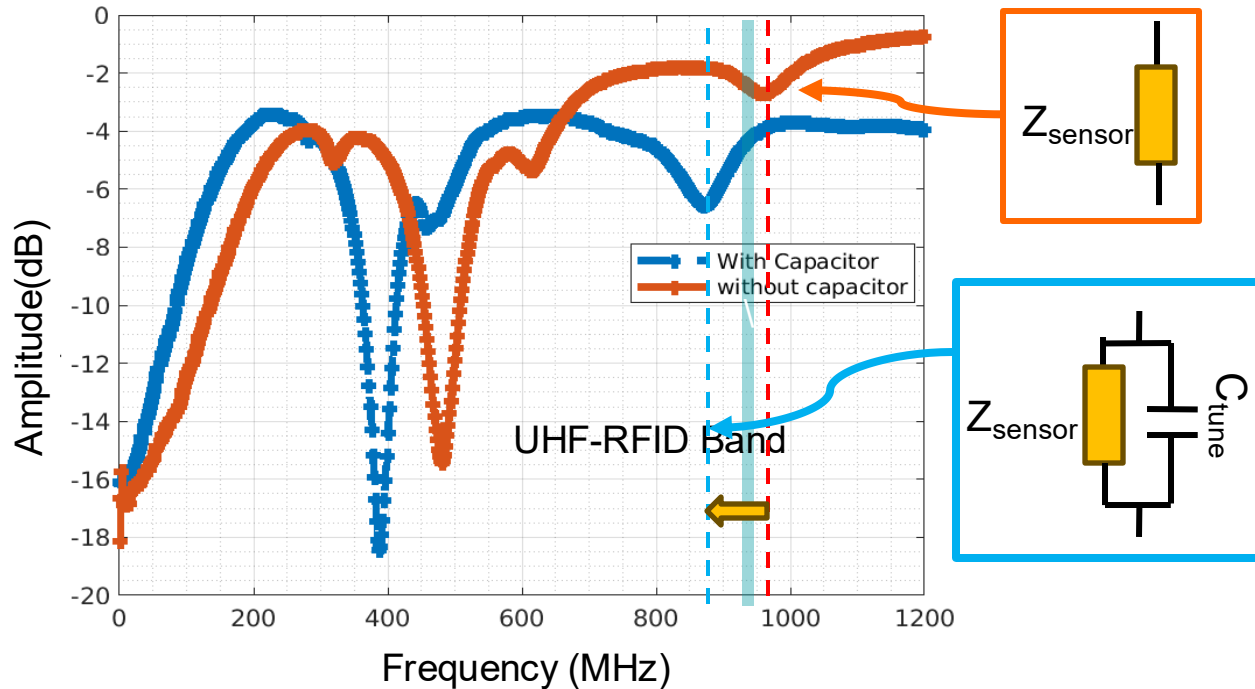
ZenseTag embeds sensor impedance into the tag digital ID.

# ZenseTag: Direct-to-RF Interface of COTS sensors



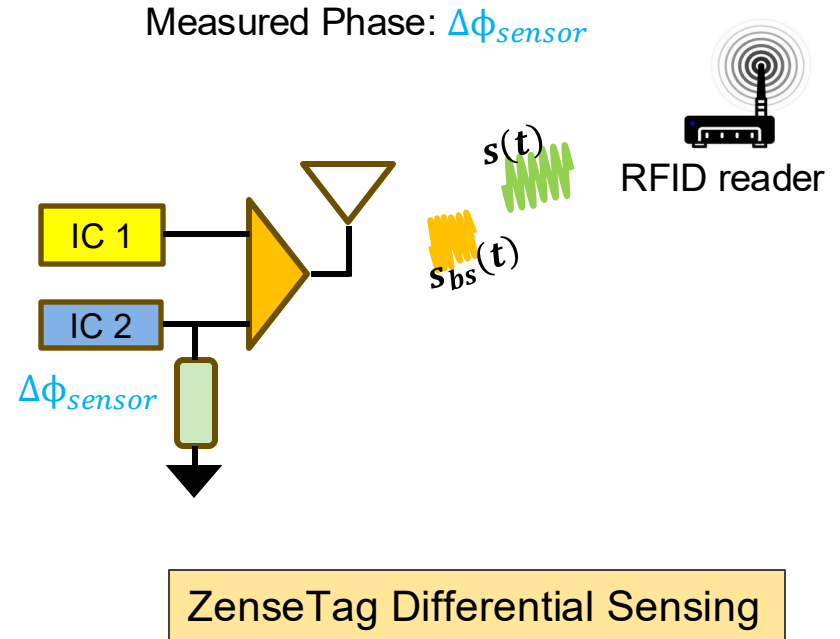
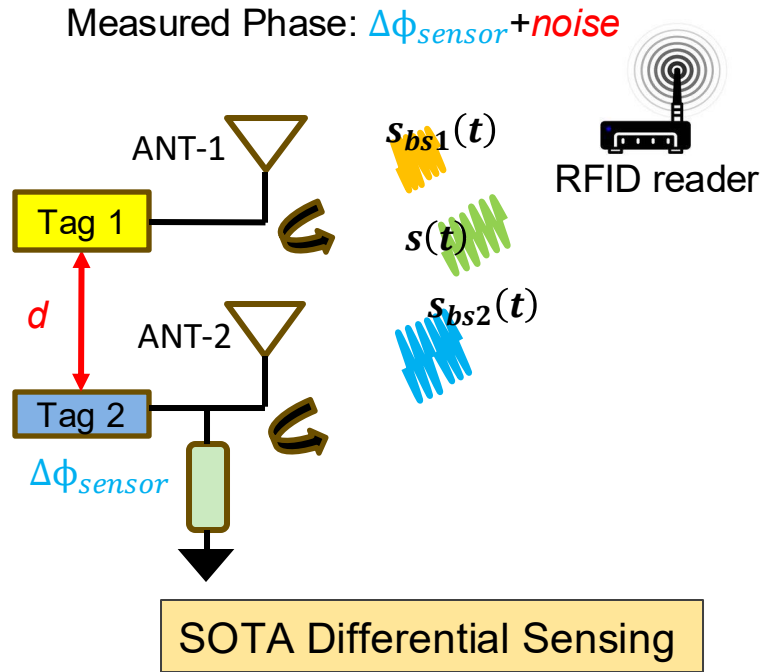
ZenseTag accurately profiles sensor impedance and tunes its resonance.

# ZenseTag: Tuning the resonance of sensors



ZenseTag tunes Sensor Resonance close to RFID band.

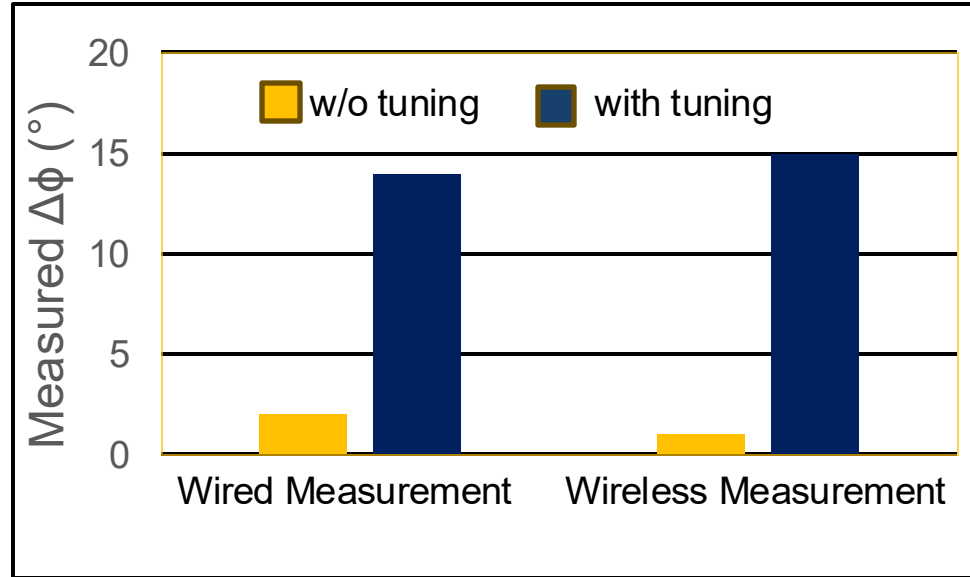
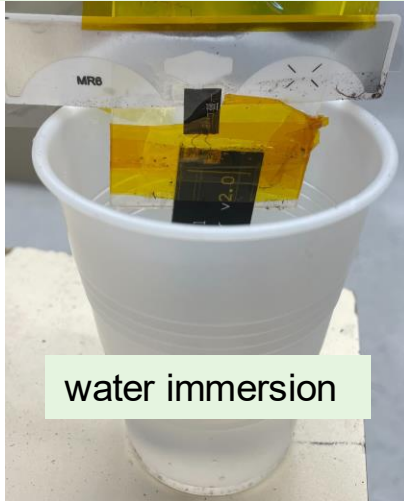
# ZenseTag: Core Contributions



ZenseTag uses a single antenna to interface 2 ICs, for robust sensing

# Benchmarks: Resonance-enhanced sensitivity

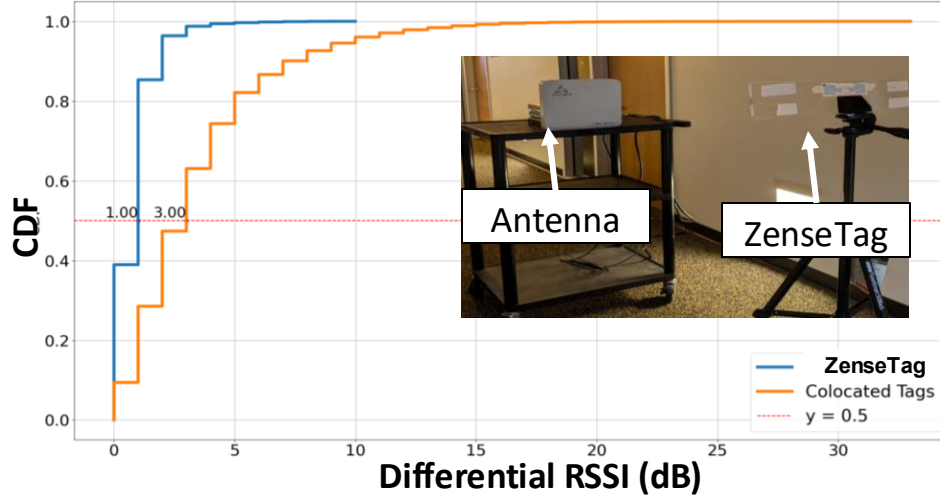
ZenseTag-Soil moisture



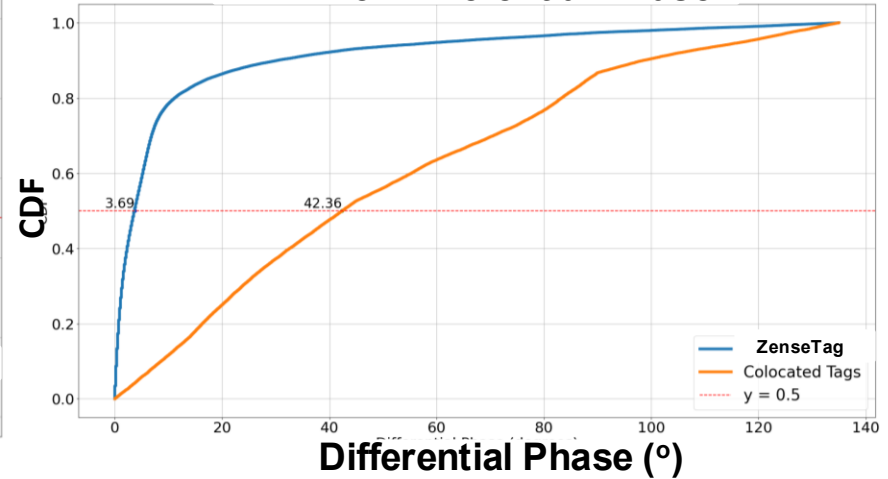
Tuning resonance achieves **7x** improvement in phase response of the sensor at RF

# Benchmarks: Multipath resilience

## CDF For Differential RSSI

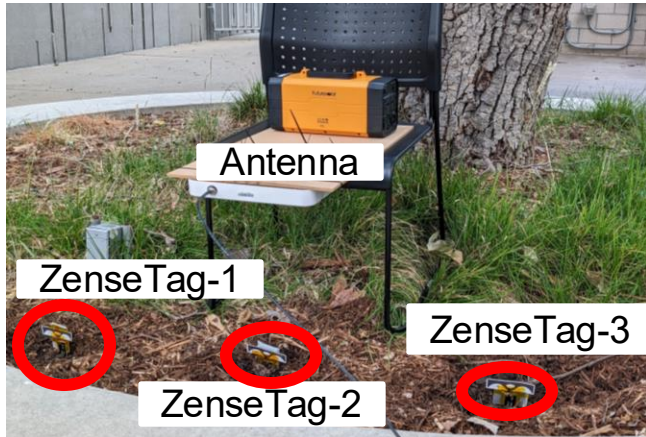


## CDF For Differential Phase

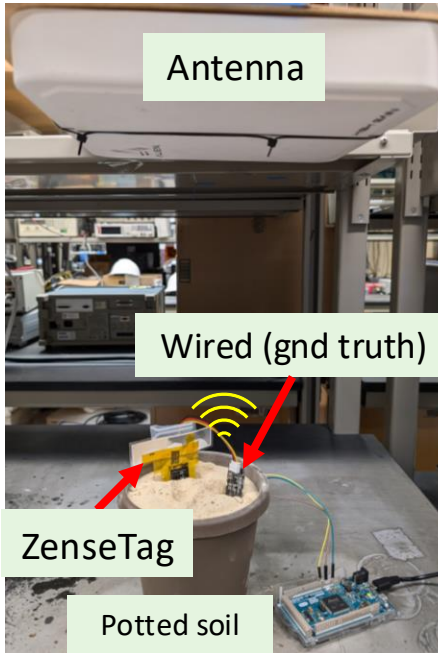


**ZenseTag: >10x accurate phase estimate, +2dB accurate amplitude estimate**

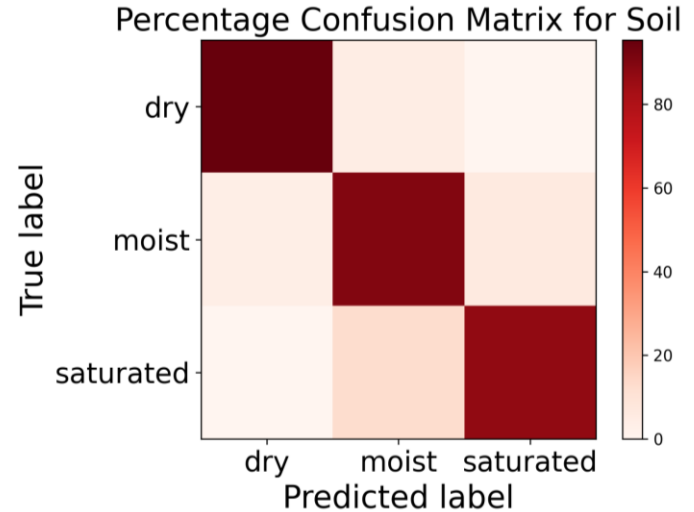
# Evaluations: Sensing soil moisture



Outdoor measurement setup



Indoor measurement setup

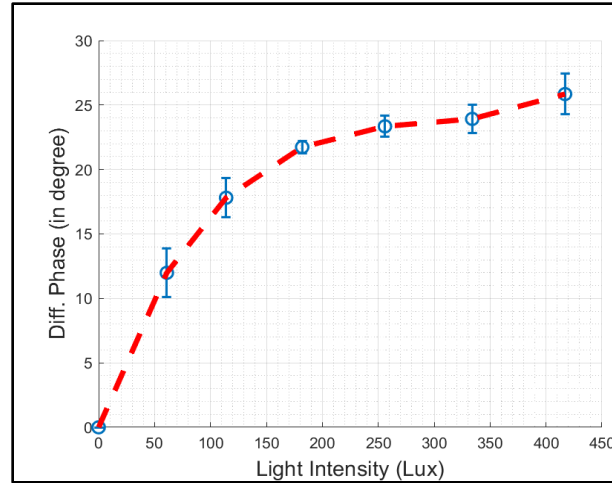


ZenseTag achieves **>93%** classification accuracy for soil moisture

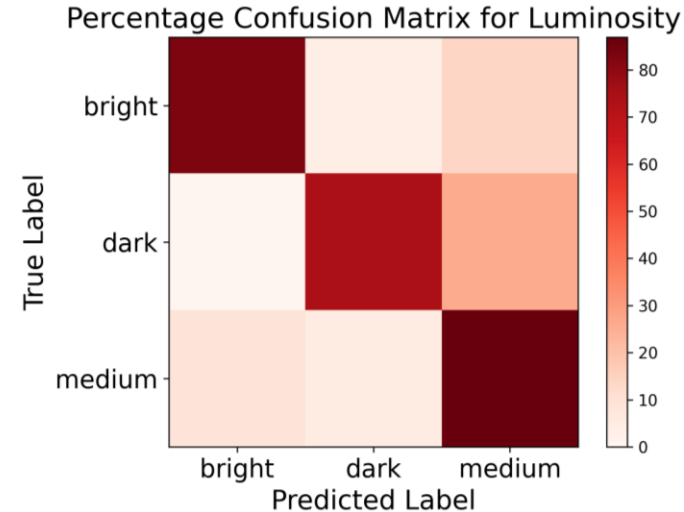
# Evaluations: Sensing Light



Measurement setup



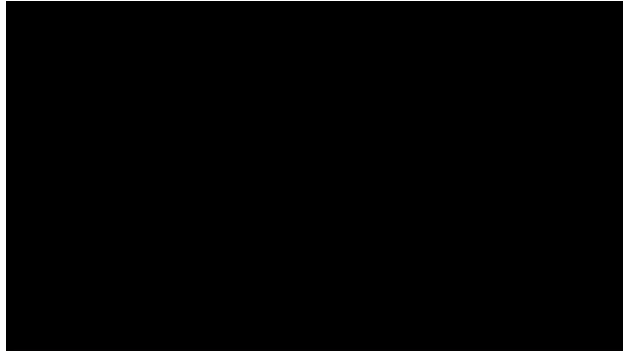
Calibration curve



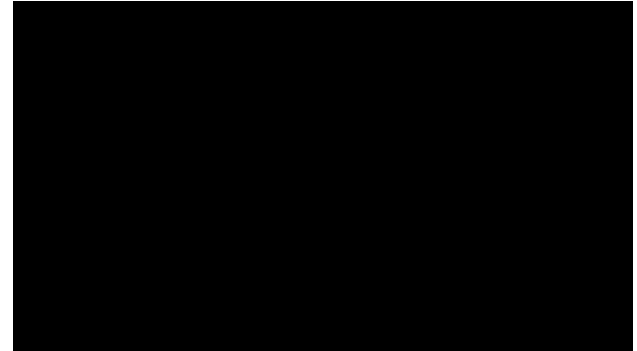
Results

ZenseTag achieves **>85%** classification accuracy for light intensity

# Evaluations: Demonstrations



ZenseTag-Luminosity  
Sensor



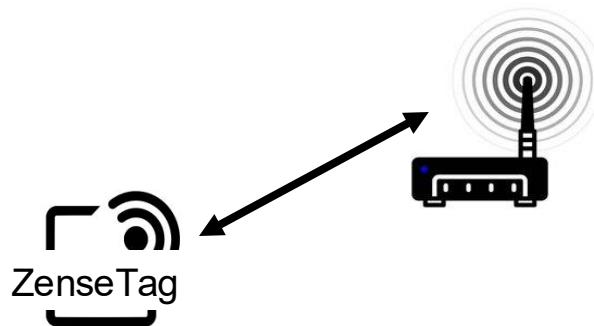
ZenseTag-Contact  
Force Sensor



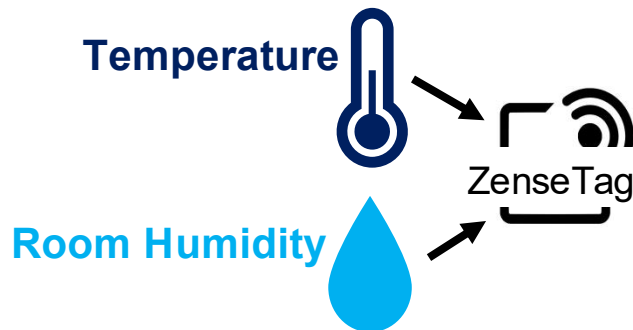
ZenseTag-Soil  
Moisture Sensor

ZenseTag enables sensing in real-time using COTS sensors

# Future Directions



Improve sensing range



Additional Sensor Interfaces



Compatibility with portable readers

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# Thank you!

Please read our paper here:

